

6.1.35. Let $A = \{a, b\}$, $B = \{1, 2\}$, and $C = \{2, 3\}$. Find each of the following sets.

- a. $A \times (B \cup C)$
- b. $(A \times B) \cup (A \times C)$
- c. $A \times (B \cap C)$
- d. $(A \times B) \cap (A \times C)$

a. $A \times (B \cup C)$

$$A = \{a, b\}$$

$$B = \{1, 2\}$$

$$C = \{2, 3\}$$

Definition of union:

$$B \cup C = \{x \in U \mid x \in B \text{ or } x \in C\}$$

$$\text{Hence, } B \cup C = \{1, 2, 3\}$$

$$\begin{aligned} A \times (B \cup C) = \{ \\ & (a, 1), (a, 2), (a, 3), \\ & (b, 1), (b, 2), (b, 3) \\ & \} \end{aligned}$$

Solution: $A \times (B \cup C) = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

b. $(A \times B) \cup (A \times C)$

$$A = \{a, b\}$$

$$B = \{1, 2\}$$

$$C = \{2, 3\}$$

$$\begin{aligned} (A \times B) = \{ \\ & (a, 1), (a, 2), \\ & (b, 1), (b, 2) \\ & \} \end{aligned}$$

$$\begin{aligned} (A \times C) = \{ \\ & (a, 2), (a, 3), \\ & (b, 2), (b, 3) \\ & \} \end{aligned}$$

Definition of union:

$$(A \times B) \cup (A \times C) = \{x \in U \mid x \in (A \times B) \text{ or } x \in (A \times C)\}$$

$$\begin{aligned} \text{Hence } (A \times B) \cup (A \times C) = \{ \\ & (a, 1), (a, 2), (b, 1), \\ & (b, 2), (a, 3), (b, 3) \\ & \} \end{aligned}$$

Solution: $(A \times B) \cup (A \times C) = \{(a, 1), (a, 2), (b, 1), (b, 2), (a, 3), (b, 3)\}$

c. $A \times (B \cap C)$

$$A = \{a, b\}$$

$$B = \{1, 2\}$$

$$C = \{2, 3\}$$

Definition of intersection:

$$B \cap C = \{x \in U \mid x \in B \text{ and } x \in C\}$$

Hence, $B \cap C = \{2\}$

$$A \times (B \cap C) = \{(a, 2), (b, 2)\}$$

Solution: $A \times (B \cap C) = \{(a, 2), (b, 2)\}$

d. $(A \times B) \cap (A \times C)$

$$A = \{a, b\}$$

$$B = \{1, 2\}$$

$$C = \{2, 3\}$$

$$(A \times B) = \{ \\ (a, 1), (a, 2), \\ (b, 1), (b, 2) \\ \}$$

$$(A \times C) = \{ \\ (a, 2), (a, 3), \\ (b, 2), (b, 3) \\ \}$$

Definition of intersection:

$$(A \times B) \cap (A \times C) = \{x \in U \mid x \in (A \times B) \text{ and } x \in (A \times C)\}$$

Hence $(A \times B) \cap (A \times C) = \{(a, 2), (b, 2)\}$

Solution: $(A \times B) \cap (A \times C) = \{(a, 2), (b, 2)\}$